Research Progress

What I learn from "Can AI Agents Agree?"

Chiharu Tomimori

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Research Motivation

Question

Can AI-agents reliably reach consensus like distributed systems?

Focus of this report

- Compare **PBFT** and **Can AI Agents Agree?**
- Identify assumptions that no longer hold
- Explore possible research directions

PBFT vs. AI-Agent Consensus

	PBFT	Can AI Agents Agree?
Node	Deterministic process	AI-Agent
Execution	Fixed protocol	Natural language reasoning
Failure Model	Byzantine node	Byzantine + reasoning errors
Output	Identical value	Probabilistic answer
Consensus Target	State / value	Answer agreement

Key Difference

PBFT assumes correct nodes always execute the protocol faithfully, while AI-agents may misunderstand the protocol itself.

Where PBFT Assumptions Break

PBFT assumes:

- Correct nodes always follow the protocol.
- Messages are interpreted correctly.
- State transitions are deterministic.

AI-agents introduce new uncertainties:

- Hallucination
- Prompt misunderstanding
- Context forgetting
- Non-deterministic reasoning

These behaviors are not covered by the classical Byzantine fault model.

Updated Research Direction

What I learned from this paper

- AI-agents are probabilistic protocol executors
- Even honest agents may fail to follow consensus protocols
- Evaluating AI-agent distributed systems is difficult

Current decision

- Pause the AI-agent research direction
- Shift my focus to Kubernetes-based distributed systems
- Explore concrete research topics in the next step

Takeaways

- PBFT and AI-agents have fundamentally different assumptions.
- Correct execution can no longer be assumed for AI-agents.
- Existing work studies answer consensus.
- Workflow consensus remains largely unexplored.
- Next step:
    - Read the system model carefully.
    - Analyze the failure assumptions.
    - Compare with PBFT formally.